Teaching and Mentoring: Bala Krishnamoorthy

When it comes to effectiveness of teaching, I aspire to achieve a level prescribed by Richard Feynman, the famous physicist. His guiding principle was that if a topic could not be explained in a freshman lecture, it was not yet fully understood. While not always feasible, I spend hours on preparing my lectures in order to make them accessible to the average undergraduate student with the minimum required background—even when teaching graduate level classes. I have won several honors and awards at my University for teaching and mentoring. I now outline the specific goals I aspire to achieve in the classroom.

Write on a tablet PC, and capture videos of lectures. Since Fall 2009, I have been writing on a tablet PC in class, whose display is projected on the screen. This set-up allows me to make increased eye contact with my students, as I am facing them rather than having my back turned to them when I write on the black/white board. I post my scribe as a PDF file on the course web page after the lecture. Mathematics is best taught by *doing* it rather than *showing* it, and the tablet PC provides a seamless way to have a record of everything I write during a lecture. I can also switch seamlessly to the demonstration of any software packages during the lecture. Furthermore, I capture my lectures using a lecture capture software, e.g., Panopto, and post the videos publicly. For a sample of scribes and videos from my lectures, see the Lecture notes on Analysis I (Math 401). I make sure in advance that my classes are assigned to rooms in which the multimedia facilities are available. I also examine the settings in the assigned classrooms before the term starts, to make sure things run smoothly.

Offer ample help outside class. I always encourage students to approach me even outside of regular check-in (i.e., office) hours. I am happy to meet with them as long as I do not have other pressing engagements. I have helped some students to do well by providing many hours of extra help, both on regular homework and on computer assignments. I count these hours as time well-spent—a brilliant student can do well with minimum assistance, but one of the main goals of teaching a course should be to make sure that *every* student understands the material.

Obtain regular feedback from students. I usually collect information regarding the backgrounds of my students at the beginning of the course—I meet with each of them individually (this is usually Problem 0 of the first homework). I make necessary changes to the structure and content of the course as demanded by the backgrounds, and on occasion, by the performance of my students in the course as it progresses. I allow my students to submit *anonymous feedback* about the class and me, at *any* time throughout the course—see the feedback form from Math 401 for a sample. Unlike the usual course evaluation forms whose collected data is not shared with me until after the term, I can make constructive changes to the current course based on such feedback so as to serve my students better. I have also used a discussion board (MS Teams chat) for some of my courses on which the students could post relevant questions or comments. Apart from giving me a timely pulse of where my students are at in the class, such a forum encourages them to engage in discussions among themselves.

I consider mentoring students an integral part of my work as a Professor. I have worked on research with 23 PhD students (21 as mentor, 17 graduated, 6 current), 10 MS students, and 16 undergraduate students. My overarching goal is to enable my students to become effective problem solvers and be fully ready for their future professional engagements. I work with students coming in with varying levels of skills (analytical, computational, oral and written communication). While some of my students have been “superstars” (they would have excelled in *any* program in the country), I get a lot of satisfaction in seeing *all* my students succeed. I am happy to work with any student who is open to taking good advice while discussing, and even challenging when appropriate, the merits of the same, and is willing to work hard. I have mentored a few students who came in with a reputation of being “hard to work with”—I grow more as a mentor when I help such students succeed! With the multi-campus system that WSU offers, I work with many students who are not physically located in my own campus. I work harder, and smarter, to build strong working relations with such students founded on mutual trust and respect. I am the Graduate Director of my department (since August 2026), and have extended my caring approach to *all* our graduate students. After all, I succeed when our students succeed.